

Edward Francisco Buccieri

ResearchGate: [Edward-Francisco-Buccieri](#) | LinkedIn: [edward-francisco-buccieri](#)

Summary & Research Interests

Graduate mathematician specializing in Optimal Transport and Theoretical Machine Learning, with applied research experience in deep learning for digital pathology. My research interests lie at the intersection of mathematical modelling, dynamical systems and machine learning.

Education

M.Sc. in Mathematics

Università degli Studi di Pavia

Date: December 2023 – July 2026

Grade: 110/110 *cum Laude*

Thesis: *Gradient Flows and Optimal Transport in Neural Network Training*

Advisor: Prof. Matteo Negri

Key Topics: Wasserstein spaces, JKO Scheme, Mean-Field Limits, Particle Gradient Flows.

B.Sc. in Mathematics

Università degli Studi di Milano-Bicocca

Date: Graduated November 2023

Thesis: *Magnetic Monopoles on Projective Spaces*

Advisor: Prof. Alessandro Tomasiello

Research Experience

Researcher

Centro di Malattie Genetiche Cardiovascolari, Policlinico San Matteo

Location: Pavia, Italy

Date: June 2024 – Present

Research Supervisor: Prof. Eloisa Arbustini

Description: Focus on the application of deep learning to digital pathology.

Conference Presentations & Published Abstracts:

- **ePoster:** *"A comparative deep learning approach using foundation models for identification of acute cellular rejection in endomyocardial biopsies of heart transplant recipients."* ESC Congress, August 2026, Munich.
- **Oral Presentation:** *"A Novel AI-Based Digital Pathology Pipeline for Acute Cellular Rejection Assessment in Endomyocardial Biopsies from heart transplanted patients."* Conoscere e Curare il Cuore (CCC), March 2026, Florence.
- **ePoster & Published Abstract:** *"AI-Based Identification of Acute Cellular Rejection in Digital Pathology Images of Endomyocardial Biopsies from heart transplanted patients."* ESC Congress, August 2025, Madrid. Abstract published in *European Heart Journal*, 46(Supplement_1), November 2025. DOI: [10.1093/eurheartj/ehaf784.4401](https://doi.org/10.1093/eurheartj/ehaf784.4401).

Technical Skills

- **Programming Languages:** Python, C, Java, MATLAB.
- **Machine Learning & Tools:** NumPy, PyTorch, TensorFlow, GeomLoss.
- **Formatting & Typesetting:** LaTeX.

Summer Schools & Workshops

- **Mathematical Foundations of Deep Learning** *Aug 3–7, 2026*
University of Cambridge, UK. Summer school in the foundational mathematics of modern AI.
- **Mathematical Foundations of Machine Learning (MFML2026)** *May 25–29, 2026*
Hotel Montana, Monte Bondone (TN), Italy. Intensive school in mathematical foundations of ML.
- **Theoretical Foundations of Machine Learning (TFML2025)** *June 23–27, 2025*
MaLGa Center, University of Genoa, Italy. Summer school in theoretical Machine Learning.

All summer schools listed above were attended on a self-funded basis.

Extracurricular Activities

- **Student Representative** *June 2024 – June 2026*
Università degli Studi di Pavia. Actively represented student interests in academic and administrative matters.
- **Volunteer Member** *Apr 2017 – May 2019*
ForumGiovani Gessate, Italy. Assisted children with homework and helped organize local events.

Language Skills

- **Italian:** Native Speaker
- **English:** CEFR C1 (TOEFL iBT 5.0/6)